Explicit low-degree square-root polynomial for primes congruent to 5 modulo 8

Kiran Kedlaya, Swastik Kopparty

doi:10.4230/LIPIcs.CCC.2024.25; CC BY 4.0

Allowed standard prerequisites (editorial).

- Polynomial vanishing sets over finite fields and formal differentiation. Author Sections 1-3; original full bibliography retained

Theorem 1.3. *Let $p \equiv 5 \mod 8$. Then there is a polynomial that computes square roots in $\mathbb{F}_p$ with degree at most $\frac{3p+1}{8}$.*

Let p be an odd prime, and let $\mathbb{F}_p$ be the finite field with p elements.

We say $f(X) \in \mathbb{F}_p[X]$ *computes square roots* in $\mathbb{F}_p$, if for all nonzero perfect squares $a \in \mathbb{F}_p$, we have:

$$f(a)^2 = a.$$

In other words, for each nonzero perfect square $a \in \mathbb{F}_p$, $f(a)$ is one of its two square roots.

Let p be a prime that is 1 mod 4. Let S be the set of nonzero perfect squares in $\mathbb{F}_p$.

In this section, we give an upper bound on the degree of polynomials computing square roots mod p, for infinitely many $p \equiv 1 \mod 4$. The upper bound is best when $p \equiv 5 \mod 8$, and we only present this case. The result and proof of this section is due to Agou, Deliglése and Nicolas [ADN03]. It remains in this paper only for completeness.

Proof. Since $p \equiv 1 \mod 4$, we get that -1 is a perfect square mod p. Let $i \in \mathbb{F}_p$ be one of the square roots of -1. Our main ingredient is the Tonelli-Shanks algorithm [Sha73, Ton91] computing square roots mod p. For $p = 4\ell + 1$, the algorithm essentially gives a formula for the square root depending on two cases. Specifically, let $u : S \to \mathbb{F}_p$ given by:

$$u(a) = \begin{cases} a^{(p+3)/8} & a^{(p-1)/4} = 1, \\ i \cdot a^{(p+3)/8} & a^{(p-1)/4} = -1. \end{cases}$$

Then for all $a \in \mathbb{F}_p$, $u(a)$ is a square root of a.

This is already quite special; the set S is partitioned into two equal sized parts S_0 and S_1, and on each S_i we have a polynomial $f_i(X)$ computing the square root of degree about $\frac{1}{2}|S_i|$. (This is the lowest possible degree, since $f_i(X)^2 - X$ is a nonzero polynomial that vanishes on all of S_i.)

Usually if we have this kind of setup, even though the f_i have unusually low degree, the unique polynomial f (obtained from the Chinese remainder theorem) which restricts to f_i on S_i has no reason to have unusually low degree. But in this case it does!

Using the usual Chinese remainder formula, we consider the polynomial $f(X) \in \mathbb{F}_p[X]$ given by:

$$\begin{aligned} f(X) &= \frac{1}{2}\left(X^{(p+3)/8}(X^{(p-1)/4}+1) - i \cdot X^{(p+3)/8}(X^{(p-1)/4}-1)\right) \\ &= \frac{1-i}{2}X^{(3p+1)/8} + \frac{1+i}{2}X^{(p+3)/8}. \end{aligned}$$

By design, we have $f(a) = u(a)$ for all $a \in S$. Finally, notice that $\deg(f) \leq \frac{3p+1}{8}$.

As a sanity check, we directly verify that $f(X)^2 \equiv X \mod (X^{(p-1)/2}-1)$. Indeed,

$$\begin{aligned} f(X)^2 &= \left(\frac{1-i}{2}\right)^2 X^{(3p+1)/4} + 2 \cdot \frac{(1-i)(1+i)}{4}X^{(4p+4)/8} + \left(\frac{1+i}{2}\right)^2 X^{(p+3)/4} \\ &= -\frac{i}{2}X^{(3p+1)/4} + X^{(p+1)/2} + \frac{i}{2}X^{(p+3)/4} \\ &= \left(-\frac{i}{2}X^{(p+3)/4} + X\right) \cdot (X^{(p-1)/2}-1) + X, \end{aligned}$$

as desired. □

References

[ADN03] Simon Joseph Agou, Marc Deléglise, and Jean-Louis Nicolas. Short polynomial representations for square roots modulo p. *Designs, Codes and Cryptography*, 28(1):33–44, 2003.

[AS97] Sanjeev Arora and Madhu Sudan. Improved low-degree testing and its applications. In *Proceedings of the twenty-ninth annual ACM symposium on Theory of computing*, pages 485–495, 1997.

[Bir00] András Biró. On polynomials over prime fields taking only two values on the multiplicative group. *Finite Fields and Their Applications*, 6(4):302–308, 2000.

[BS96] Eric Bach and Jeffrey Outlaw Shallit. *Algorithmic number theory: Efficient algorithms*, volume 1. MIT press, 1996.

[Car11] NA Carella. Formulas for the square root modulo p. *arXiv preprint arXiv:1101.4605*, 2011.

[CKL15] Seunghwan Chang, Bihtnara Kim, and Hyang-Sook Lee. Polynomial representations for n-th roots in finite fields. *Journal of the Korean Mathematical Society*, 52(1):209–224, 2015.

[CS00] Don Coppersmith and Igor Shparlinski. On polynomial approximation of the discrete logarithm and the diffie—hellman mapping. *Journal of Cryptology*, 13:339–360, 2000.

[GR05] Venkatesan Guruswami and Atri Rudra. Limits to list decoding reed-solomon codes. In *Proceedings of the thirty-seventh annual ACM symposium on Theory of computing*, pages 602–609, 2005.

[GS98] Venkatesan Guruswami and Madhu Sudan. Improved decoding of reed-solomon and algebraic-geometric codes. In *Proceedings 39th Annual Symposium on Foundations of Computer Science (Cat. No. 98CB36280)*, pages 28–37. IEEE, 1998.

[Kal91] Erich Kaltofen. Effective noether irreducibility forms and applications. In *Proceedings of the twenty-third annual ACM symposium on Theory of Computing*, pages 54–63, 1991.

[Mas84] Richard Clive Mason. *Diophantine equations over function fields*, volume 96. Cambridge University Press, 1984.

[MW86] Gary Mullen and David White. A polynomial representation for logarithms in gf (q). *Acta arithmetica*, 3(47):255–261, 1986.

[Sch06] Wolfgang M Schmidt. *Equations over finite fields: an elementary approach*, volume 536. Springer, 2006.

[Sha73] Daniel Shanks. Five number-theoretic algorithms. In *Proceedings of the Second Manitoba Conference on Numerical Mathematics (Winnipeg), 1973*, 1973.

[Sho09] Victor Shoup. *A computational introduction to number theory and algebra*. Cambridge university press, 2009.

[Sto81] W Wilson Stothers. Polynomial identities and hauptmoduln. *The Quarterly Journal of Mathematics*, 32(3):349–370, 1981.

[Ton91] Alberto Tonelli. Bemerkung über die auflösung quadratischer congruenzen. *Nachrichten von der Königl. Gesellschaft der Wissenschaften und der Georg-Augusts-Universität zu Göttingen*, 1891:344–346, 1891.

[VZGG13] Joachim Von Zur Gathen and Jürgen Gerhard. *Modern computer algebra*. Cambridge university press, 2013.

[WB83] Lloyd R Welch and Elwyn R Berlekamp. Error correction for algebraic block codes. *US patent*, (4,633,470), 1983.

[Win02] Arne Winterhof. Polynomial interpolation of the discrete logarithm. *Designs, Codes and Cryptography*, 25(1):63–72, 2002.